

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (E & C)

Sem - IV University Examination, 2010-11

EC - 210 Analog Communication

Date: 13.05.11, Friday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question Number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION - I

- Q: 1 (a) Explain the concept of modulation using frequency shifting property of Fourier transforms. (3)
- (b) Find the Fourier transforms.
- (i) $t^n e^{-at} u(t)$ (4)
- (ii) $e^{-at} \sin \omega_0 t u(t)$, where $a > 0$
- (c) Explain diagonal peak clipping in the diode envelope detector. (4)

- Q: 2 (a) Explain the circuit of BJT collector modulator class C amplifier for AM generation. (6)
- (b) A standard AM transmission, sinusoidally modulated to a depth to 30%, produces side frequency of 4.928 & 4.914 MHz The amplitude of each side frequency is 75 volt. Determine the amplitude & frequency of carrier. (2)
- (c) Compare FM over AM. The rms antenna current of AM radio transmitter is 10A & when unmodulated & 12A when sinusoidally modulated. Calculate modulation index. (4)

OR

- Q: 2 (a) What is modulation? Explain the need for modulation. (3)
- (b) Explain the third method of generating an SSBSC signal. (6)
- (c) Certain AM transmitter radiates 10 kw with the carrier unmodulated and 25kw when the carrier is sinusoidally modulated .calculate the modulation index, if another wave corresponding to 70% modulation is transmitted simultaneously ,determine the total radiated power. (3)

- Q: 3 (a) Explain pilot carrier SSB. (2)
- (b) Explain FET singly balanced modulator circuit. (6)
- (c) What do you mean by signal to noise ratio? A mixer circuit having a noise figure of 20 dB is preceded by an amplifier having a noise figure of 9 dB and an available power gain of 15dB. What is the overall noise figure of combinations? (4)

OR

- Q: 3 (a) Prove that the ratio of anyone side frequency to the total power transmitted is 1/6. (3)
- (b) Compare DSB-SC over SSB-SC. (2)

- (c) Show that for a power signal, the PSD $S_g(\omega)$ is the Fourier transform of the auto correlation function $R_g(t)$. (4)
- (d) An AM wave has maximum and minimum amplitudes are measure as 7.2v and 1.8v respectively the modulating and carrier signal having frequency of 400 Hz and 1 MHz respectively. Calculate (i) modulation index (ii) transmission efficiency (iii) bandwidth occupied by such wave. (3)

SECTION - II

- Q: 4 (a) Define noise factor and derived friis's formula. (3)
- (b) Explain how inductance & capacitance control the noise circuit. Why thermal noise is called white noise? (4)
- (c) The Equivalent noise resistance for an amplifier is 300Ω & the equivalent shot noise current is $5 \mu A$ the amplifier is fed from a 150Ω , $10\mu V$ rms sinusoidal signal source. Calculate the individual noise voltages at the input and the input signal to noise ratio in dB. The noise bandwidth is 10MHz. (4)

- Q: 5 (a) Explain how FM signals can be detected by basic foster seely discriminator. (4)
- (b) Write note on: Varactor diode frequency modulator. (4)
- (c) An angle modulated signal with carrier frequency $\omega_c = 2\pi \times 10^4$ is described by the equation (4)
- $$V(t) = 5 \cos(\omega_c t + 5 \sin 2000\pi t)$$
- Find the power of the modulated signal.
 - Find the deviation ratio.
 - Find the bandwidth of the signal.

OR

- Q: 5 (a) Explain Pre-emphasis and de-emphasis. (4)
- (b) Derive PM from FM. (4)
- (c) Define: (4)
- Modulation index.
 - Carson's Rule.
 - Narrowband FM.
 - Deviation Ratio.

- Q: 6 (a) Explain super heterodyne receivers and super heterodyne principle. (6)
- (b) Define sensitivity, selectivity, double spotting. (3)
- (c) An AM broadcast receiver has an IF of 465 KHz and is tuned to 1000 KHz, and the RF stage has one tuned circuit with Q of 50. (3)
- Find the image frequency.
 - Find the image rejection in dB

OR

- Q: 6 (a) Explain automatic gain control in details. (4)
- (b) Discuss the application & function of amplitude limiters in frequency modulator. (5)
- (c) The broad cast receiver is tuned to a signal circuit at 950 KHz. Find all the unwanted signal frequencies for harmonics up to the second. Which of these fall within 200 KHz of the desired frequency. Take IF=455 KHz (3)